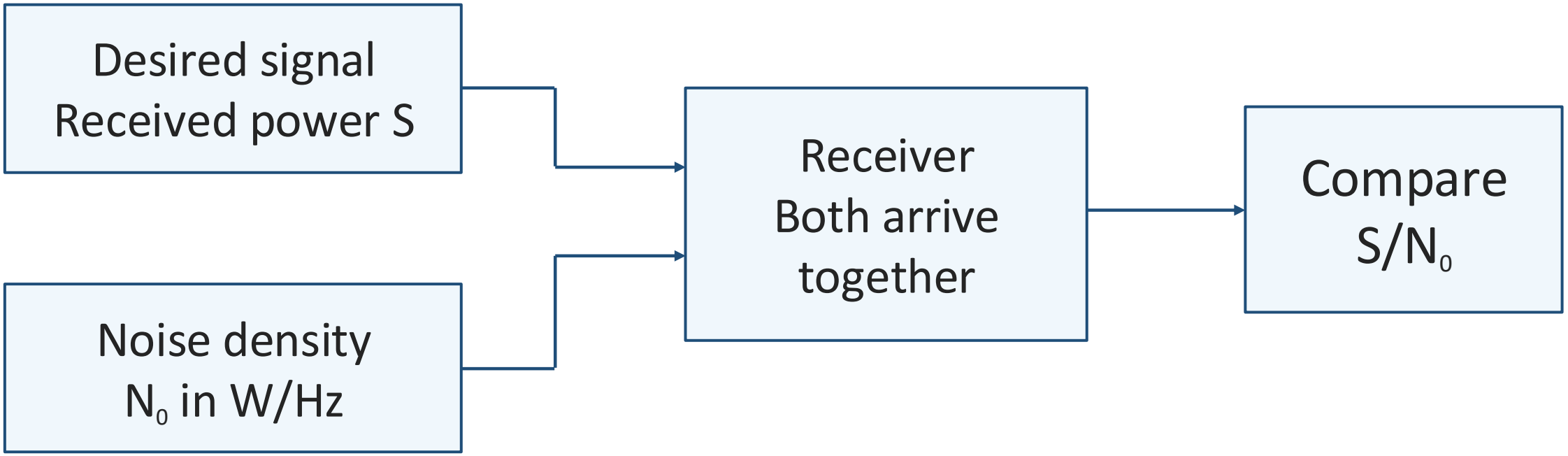


# Spacecraft Communications
# ENAE 694
# Noise and the Complete Link Budget

Dave Israel (disrael@umd.edu)

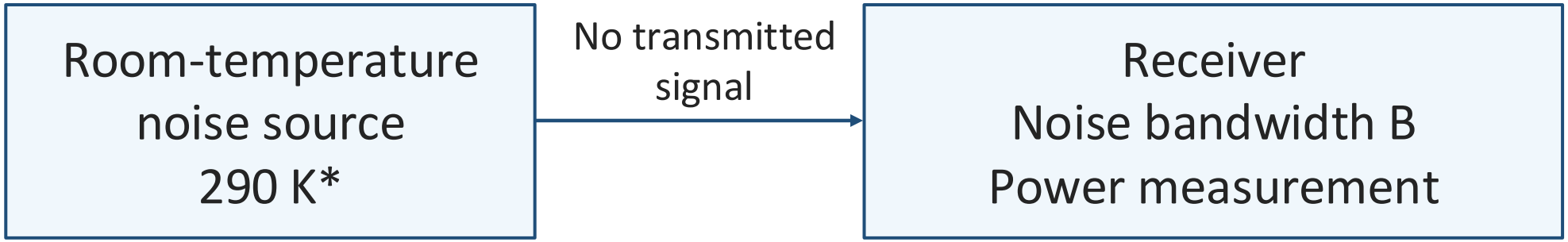
October 5, 2026

# Received Power Is Only Half the Story




$S/N_0$ compares received signal power with noise per hertz.

# Question: Will the Receiver Read Zero?

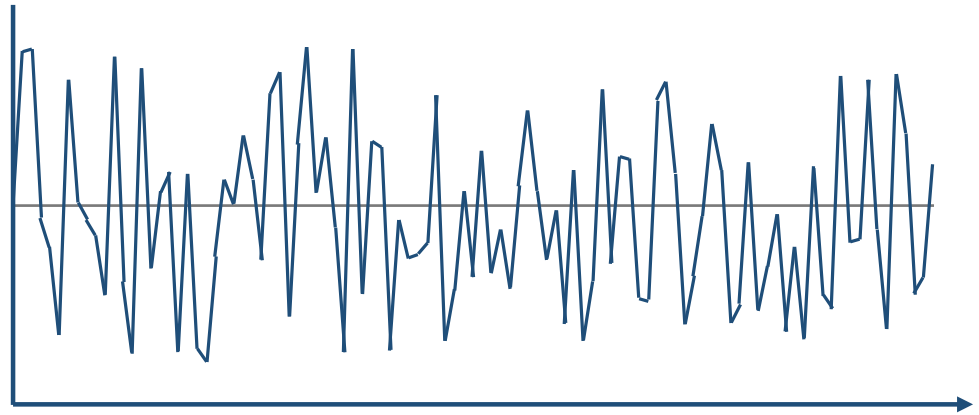



Thermal motion creates random electrical fluctuations.
What power reaches the receiver?

* Matched connection: the receiver accepts the source's available noise power.

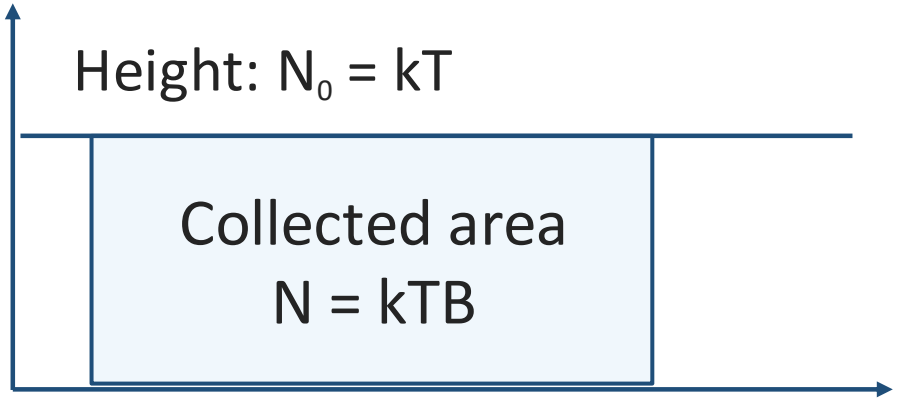
# Random Fluctuations Carry Real Power

Noise voltage versus time



No repeating cycle.
Mean voltage is approximately zero.

Noise density versus frequency




Width: B hertz

The squared fluctuations have a positive average: the noise power is real.

# Thermal Noise Density: $N_0 = kT$

Thermal noise has density $N_0 = kT$, in W/Hz.

$[N_0] = -228.6 + 10 \log_{10}(T/K)$, in dBW/Hz.

At 290 K: $[N_0] = -204.0$ dBW/Hz $= -174.0$ dBm/Hz.

Doubling T raises the noise density by 3.0 dB.

Total noise across bandwidth B is $N = N_0B = kTB$.

# Noise Figure: The Component Specification

A component adds noise and degrades the input SNR.

Noise factor measures that degradation: $F = SNR_{in}/SNR_{out}$.*

Noise figure expresses it in dB: $NF = 10 \log_{10} F$.

Lower NF means less added noise under the same conditions.

Component noise figure and gain describe each receiver stage.

* NF uses a source at $T_o = 290$ K and the same noise bandwidth at input and output.

# Three Different Meanings of Temperature

| | | |
|---|---|---|
| Physical temperature $T_{physical}$ | Actual hardware temperature. A thermometer could measure it. | Example: 290 K |
| Equivalent noise temperature $T_e$ | Receiver-added noise expressed as an equivalent input temperature. | Example: 50 K |
| Reference temperature $T_0$ | The fixed convention for defining noise factor and noise figure. | Always 290 K |

One receiver can have 290 K hardware, 50 K equivalent noise, and a 290 K reference.

# A Useful Way to Describe Added Noise

1. Noise arrives

   The receiver amplifies the incoming noise.

2. The receiver adds noise

   The output contains both contributions.

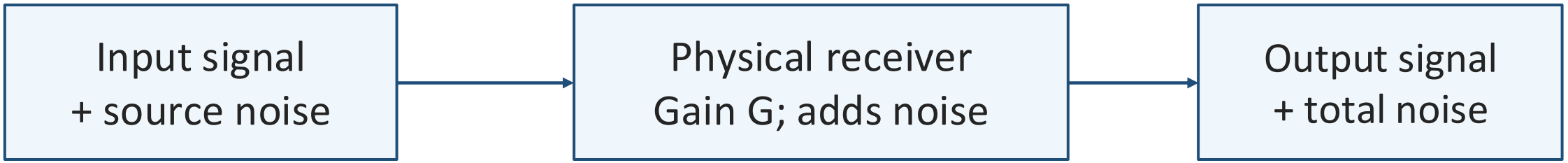
3. Express that addition at the input

   $T_e$ tells us the strength of an equivalent thermal source.

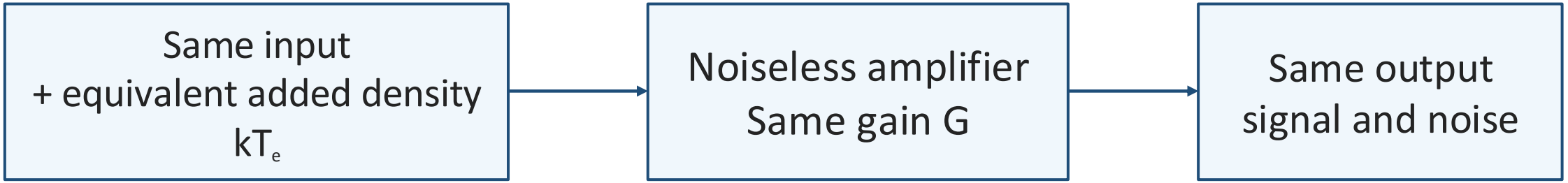
NF = 0.69 dB gives $T_e = 290(10^{NF/10} - 1) \approx 50$ K.

# Two Models, Exactly the Same Output

## ACTUAL RECEIVER




## EQUIVALENT CALCULATION MODEL




We relocate added noise in the model. The hardware and its output are unchanged.

# Which Noise Contribution Matters Most?

Same receiver: NF = 0.69 dB.
Convert once: $T_e \approx 50$ K for the system-noise sum.*

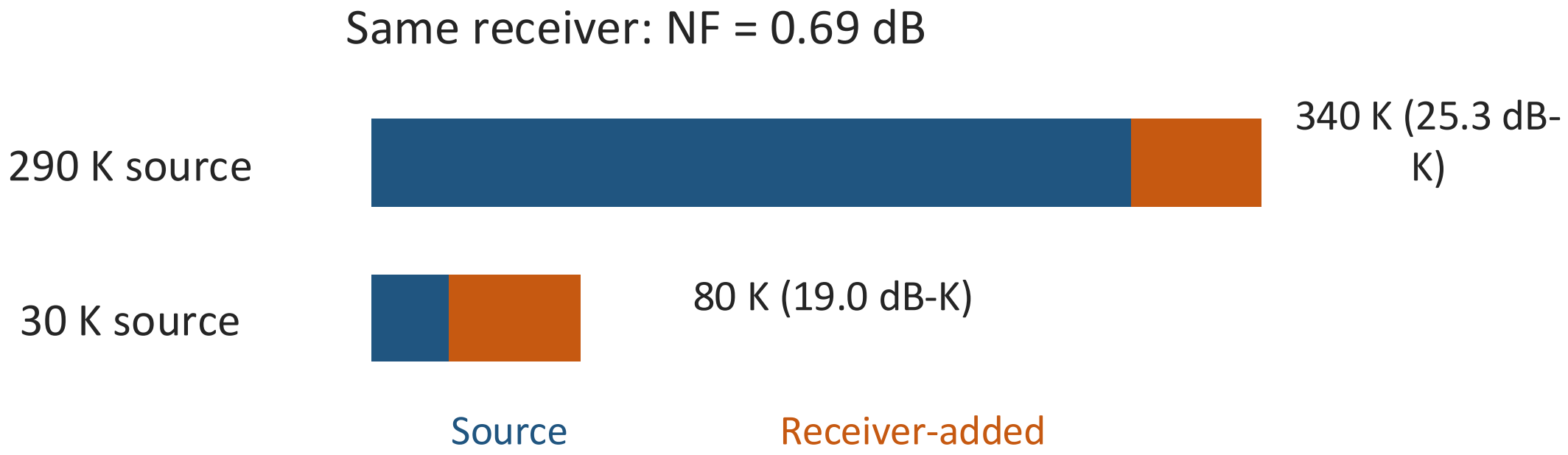
Cold-sky source: 30 K
$T_{sys} = 80$ K (19.0 dB-K)
Receiver noise is the larger part.

Room-temperature source: 290 K
$T_{sys} = 340$ K (25.3 dB-K)
Source noise is the larger part.

Lowering receiver noise helps most when receiver noise dominates.

* Equivalent-input model: JPL DESCANSO, Vol. 10, Ch. 2. The added source is a calculation model.

# Antenna Noise Need Not Be 290 K




Lower antenna noise gives lower total noise with the same receiver NF.

# Antenna Noise: The Practical Picture

The antenna collects noise as well as the wanted signal.

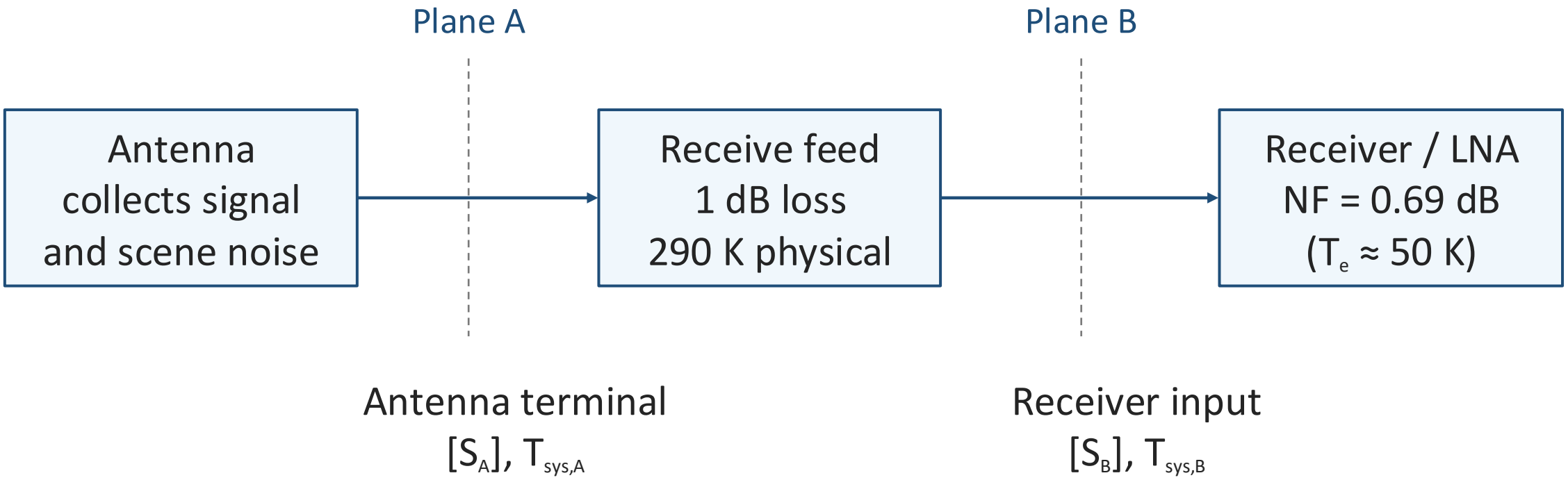
Sky, atmosphere, ground, and the Moon can change $T_A$.

The antenna pattern and pointing determine how much enters.

Use a specified antenna temperature for the link calculation.

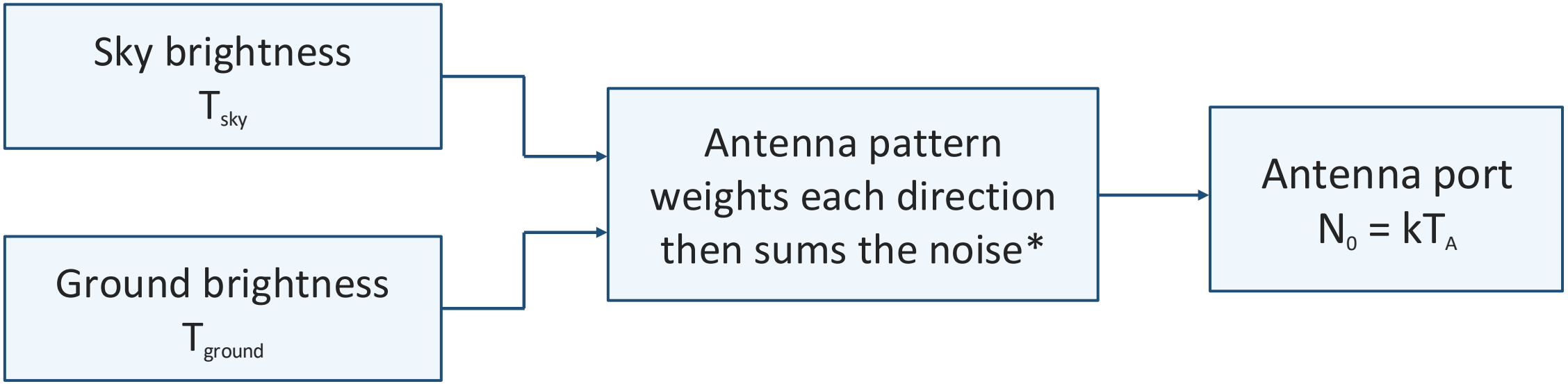
Keep external antenna noise separate from receiver noise figure.

# Reference Planes in the Receiving System




Use signal and total equivalent noise at the same plane.

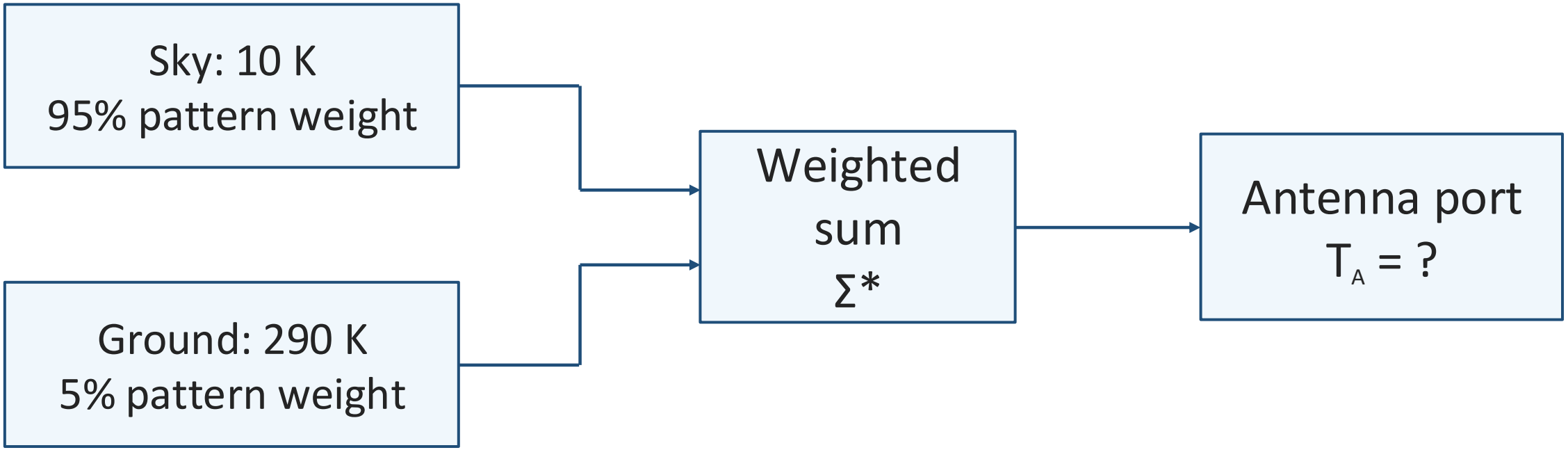
# The Pattern Weights the Whole Scene




Sky temperature describes radiation from a direction.
Antenna temperature describes the noise delivered at the port.

* Lossless-antenna model. Source: Condon & Ransom, Essential Radio Astronomy, Ch. 3.

# Question: How Much Does Ground Matter?




* Lossless antenna. Weights represent integrated antenna response in each region.

# Worked Answer: The Weighted Noise

$T_A = \Sigma\, w_i T_i$ (lossless antenna)

Sky: 0.95 × 10



9.5 K

Ground: 0.05 × 290

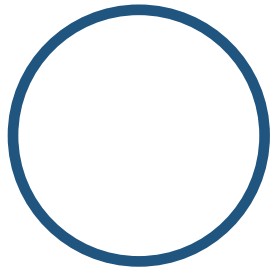


14.5 K

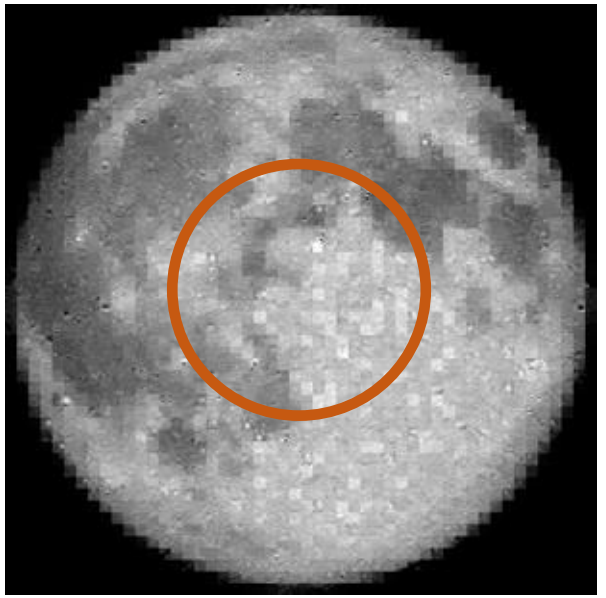
$T_A = 0.95(10) + 0.05(290) = 24.0$ K

The 5% ground pickup contributes more noise than the 95% sky pickup.

# The Moon and Antenna Beam Coupling







OFF the Lunar Disk

Assume $T_A$ = 30 K*

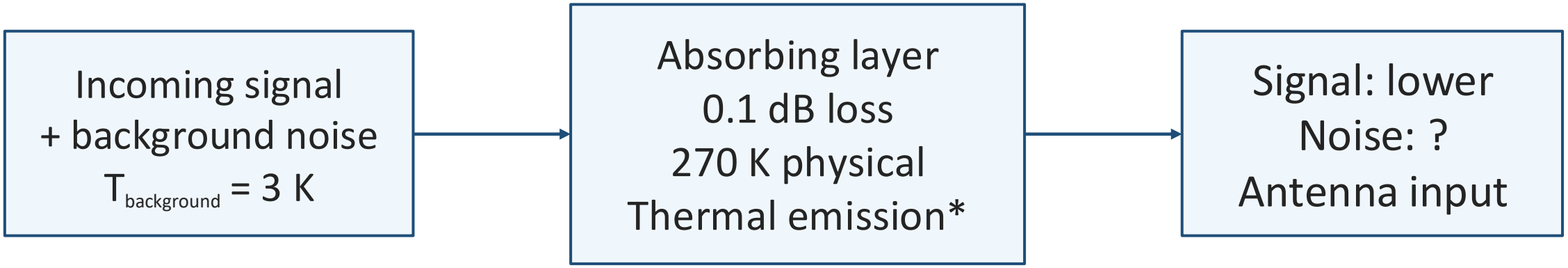
ON the Lunar Disk

Assume $T_A$ = 230 K

* Assumed scene temperatures. Beam outlines are schematic. Moon image: NASA SVS / LRO.

Atmosphere at 270 K

# Question: What Reaches the Antenna?



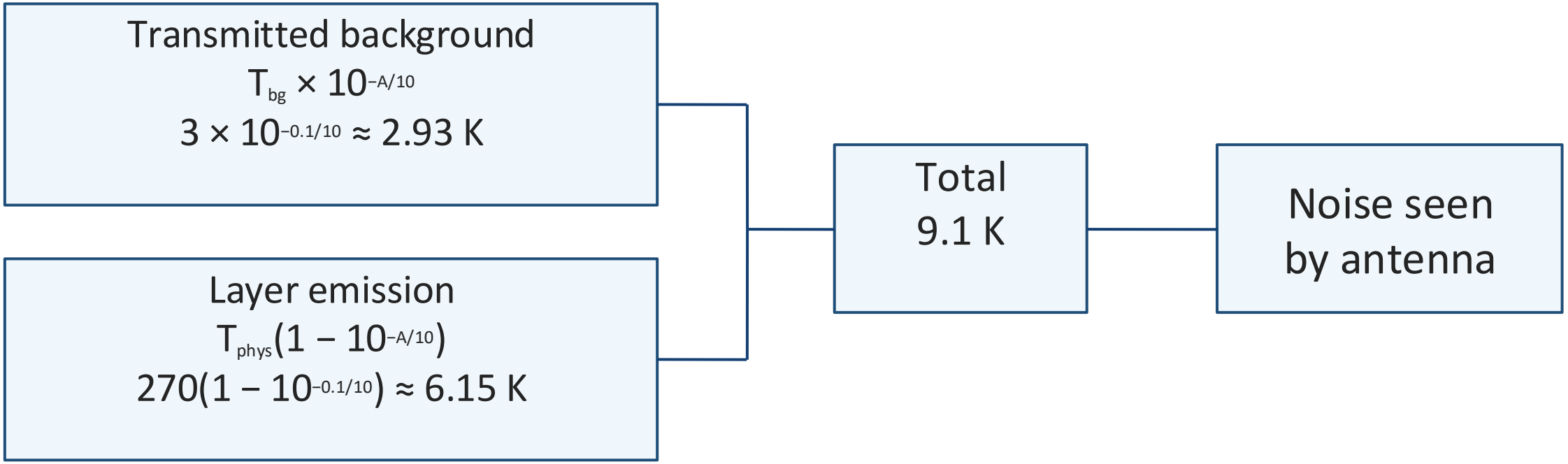

The layer attenuates incoming radiation and generates its own thermal noise.

* Isothermal layer model. Sources: NRAO Essential Radio Astronomy, Ch. 1; DSN 810-005.

Atmosphere at 270 K

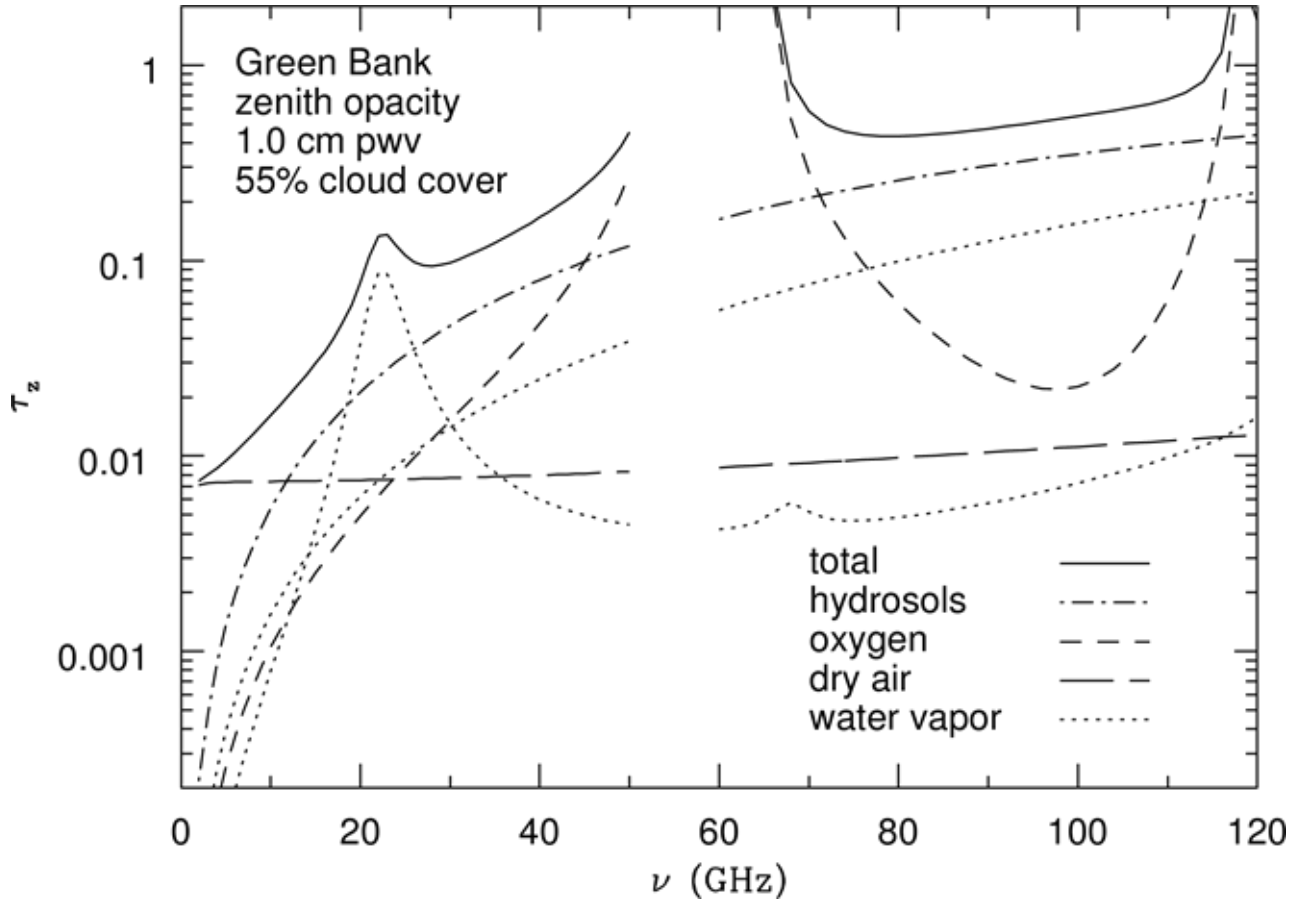
# Worked Answer: Loss and Emission

Signal ledger: $[S_{out}] = [S_{in}] - 0.1$ dB




$$T_{seen} = T_{bg}10^{-A/10} + T_{phys}(1 - 10^{-A/10}) \approx 9.1 \text{ K}.$$

# Weather and Frequency Atmospheric Absorption



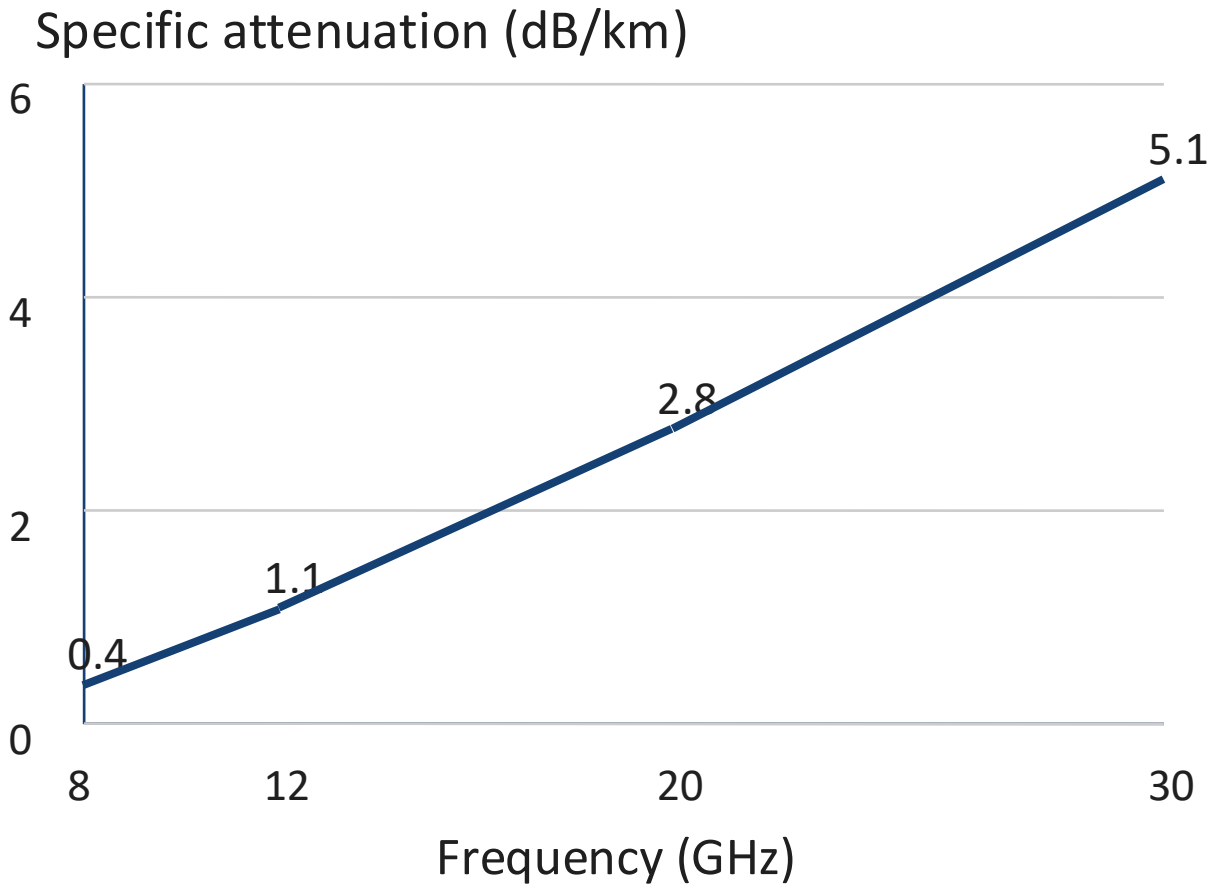

Water vapor has a feature near 22 GHz.
Oxygen absorbs strongly around 60 GHz.

Cloud liquid water adds loss.
Dry, high sites reduce water along the atmospheric path.

This curve is gas/cloud opacity, not rain attenuation.*

* Green Bank summer example. Attenuation in dB = 4.3 × opacity τ. Source: Condon & Ransom, Fig. 1.2.

# Weather and Frequency
# Rain Attenuation




Same rain rate: 25 mm/h.*

For these microwave bands, higher frequency gives much more rain attenuation.

Total fade also depends on rain-path length and elevation. Ka-band generally needs more rain margin than X-band.

* ITU-R P.838-3, Table 5; horizontal polarization. Connected model samples, not station fade predictions.

# Weather and Ground Stations
# Rain Margin and Site Selection

Rain margin is extra allowance for weather fades at a chosen availability.
Choose it using frequency, local rain statistics, elevation, and path geometry.
Dry climates and high sites often reduce atmospheric attenuation and noise.
Higher elevation shortens the path through the lower atmosphere.
Site diversity can keep service available when rain affects one station.*
Weather reduces S and can raise antenna noise, lowering G/T.

* ITU-R P.618-14. Required $S/N_o$ and adequacy come later; no universal rain margin is assigned here.

# What Attenuates Signal? What Adds Noise?

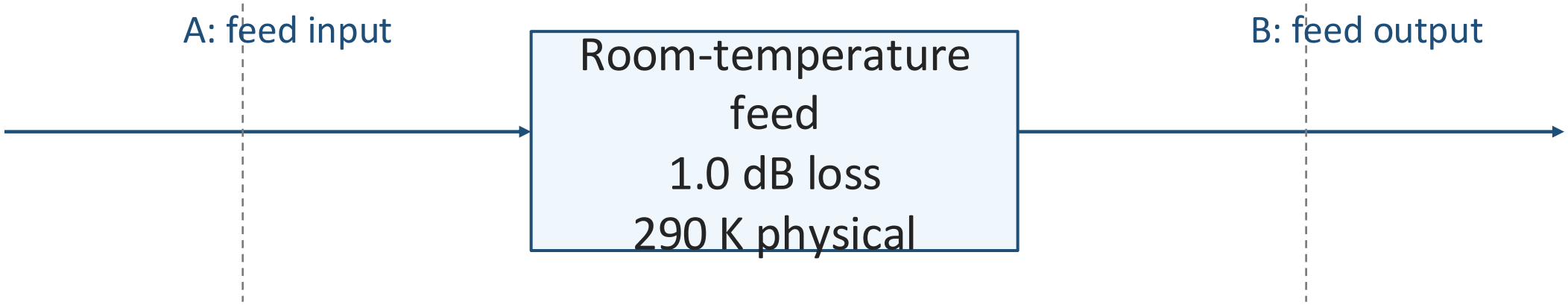
| Mechanism | Signal | Noise |
|---|---|---|
| Free space, pointing, polarization mismatch | Less received power | No thermal emission implied by this loss |
| Room-temperature feed / absorbing atmosphere | Attenuated | Incoming noise attenuated; thermal emission added |
| Receiver amplifier | Amplified | Incoming noise amplified; receiver noise added |

# Break

# Feed Loss and Noise at Two Planes

Antenna noise: 30 K
Signal: $[S_A]$

Signal: $[S_A]$ − 1 dB
Antenna noise: 23.8 K




Feed noise referred to input A:
$(L_f - 1)T_{physical} = 75.1$ K

Feed noise delivered at output B:
$(1 - 1/L_f)T_{physical} = 59.6$ K

$L_f = 10^{[L_f]/10}$. Both terms represent the same feed noise at different planes.

Lunar Link

# One System, Two Noise Ledgers

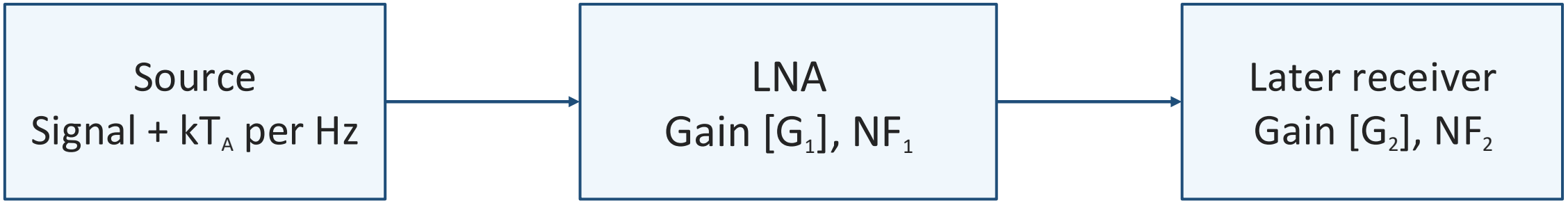
$T_{sys,A} = T_A + (L_f - 1)T_{phys} + L_f T_{e,rx}$;  $T_{sys,B} = T_{sys,A}/L_f$

| Contribution | At A: before feed | At B: after feed |
|---|---|---|
| Antenna | 30.0 K | 23.8 K |
| Feed emission | 75.1 K | 59.6 K |
| Receiver | 63.0 K | 50.0 K |
| $T_{sys}$ | 168.0 K | 133.5 K |
| $[T_{sys}]$ | 22.3 dB-K | 21.3 dB-K |

A to B: [S] and $[T_{sys}]$ both fall 1.0 dB. $[S/N_0]$ stays unchanged.

# Where Each Stage Adds Noise

Source noise passes through both stages.
Each stage also adds its own noise.




The LNA boosts the signal before later stages add their noise.
That makes later-stage noise less important relative to the signal.

Input referral expresses all added noise at one common reference plane.

# The Model: Friis Noise Equations

1. Convert catalog values: $F = 10^{NF/10}$, $G = 10^{[G]/10}$.

2. Combine the stages using their linear factors:

$$F_{total} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1 G_2) + \ldots$$

3. Return to dB: $NF_{total} = 10 \log_{10} F_{total}$.

The first stage enters directly. Early gain reduces later terms.

# Question: How Much Early Gain Helps?

LNA NF = 0.68 dB. Following receiver NF = 6.48 dB.

Keep both noise figures fixed as LNA gain changes.

Calculate combined NF for LNA gains of 10, 20, and 30 dB.

How much improvement does each extra 10 dB provide?

What sets the limit as LNA gain becomes very large?

# Worked Answer: 10, 20, and 30 dB Gain

$F_i = 10^{NF_i/10}$; $G_1 = 10^{[G_1]/10}$; $NF_{total} = 10 \log_{10} F_{total}$.

$F_{total} = F_1 + (F_2 - 1)/G_1 \approx 1.17 + 3.45/G_1$.

10 dB: $G_1 = 10$; $NF_{total} \approx 1.8$ dB.

20 dB: $G_1 = 100$; $NF_{total} \approx 0.80$ dB.

30 dB: $G_1 = 1000$; $NF_{total} \approx 0.69$ dB.

The limit is the first LNA's NF ≈ 0.68 dB.

# Receiver Cascade
# From Component noise figure to the Link Budget

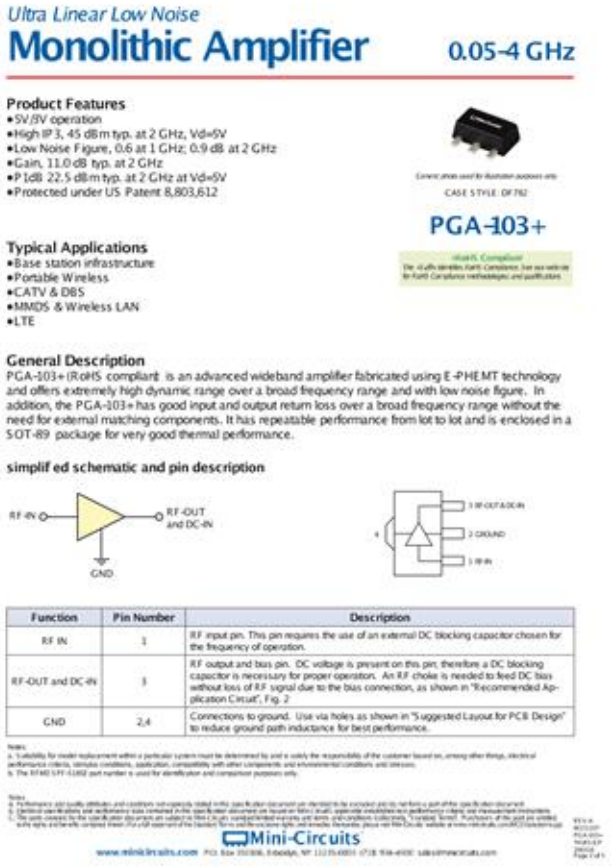



Cascade NF ≈ 0.69 dB.

Convert once for G/T: $T_{e,rx} = 290(F_{total} - 1) \approx 50$ K.

NF is the component figure of merit; kelvins combine receiver and antenna noise.

# Receiver Components
# Reading an LNA Datasheet



*Ultra Linear Low Noise*

**Monolithic Amplifier** 0.05-4 GHz

**Product Features**
- 5V,8V operation
- High IP 3, 45 dBm typ. at 2 GHz, Vd=6V
- Low Noise Figure, 0.6 at 1 GHz; 0.9 dB at 2 GHz
- Gain, 11.0 dB typ. at 2 GHz
- P1dB 22.5 dBm typ. at 2 GHz at Vd=6V
- Protected under US Patent 8,803,612

CASE STYLE: DF762

**PGA-103+**

**Typical Applications**
- Base station infrastructure
- Portable Wireless
- CATV & DBS
- MMDS & Wireless LAN
- LTE

**General Description**

PGA-103+ (RoHS compliant) is an advanced wideband amplifier fabricated using E-PHEMT technology and offers extremely high dynamic range over a broad frequency range and with low noise figure. In addition, the PGA-103+ has good input and output return loss over a broad frequency range without the need for external matching components. It has repeatable performance from lot to lot and is enclosed in a SOT-89 package for very good thermal performance.

**simplif ed schematic and pin description**

RF-IN · RF-OUT and DC-IN · GND

| Function | Pin Number | Description |
|---|---|---|
| RF IN | 1 | RF input pin. This pin requires the use of an external DC blocking capacitor chosen for the frequency of operation. |
| RF-OUT and DC-IN | 3 | RF output and bias pin. DC voltage is present on this pin; therefore a DC blocking capacitor is necessary for proper operation. An RF choke is needed to feed DC bias without loss of RF signal due to the bias connection, as shown in "Recommended Application Circuit", Fig. 2 |
| GND | 2,4 | Connections to ground. Use via holes as shown in "Suggested Layout for PCB Design" to reduce ground path inductance for best performance. |

Mini-Circuits

The catalog gives NF and gain at specified frequencies and bias.

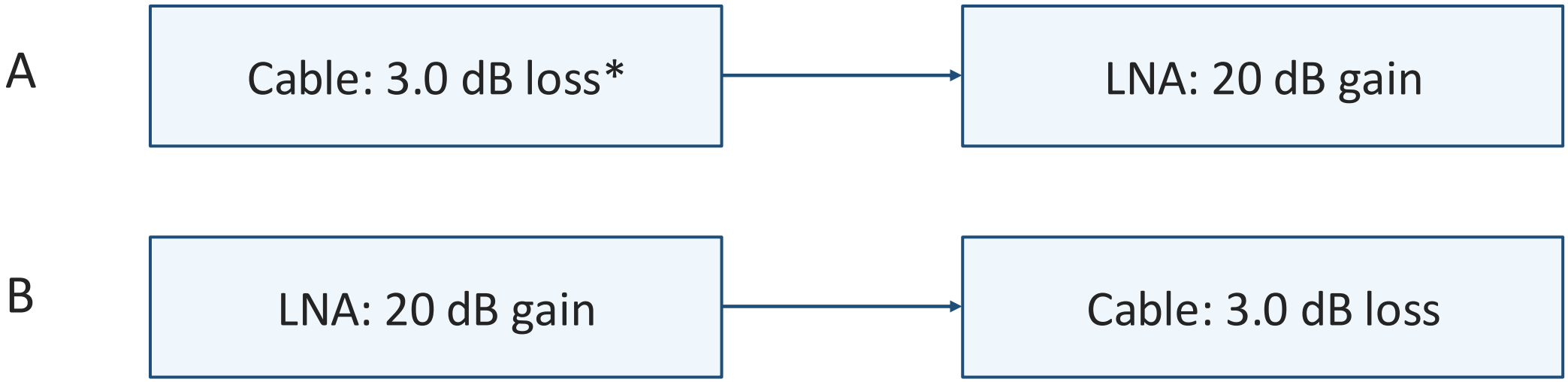
This PGA-103+ example lists:
NF: 0.6 dB at 1 GHz
NF: 0.9 dB at 2 GHz
Gain: 11 dB at 2 GHz

Also check power, matching, linearity, and operating band.*

* Catalog-reading example, not the 8.4 GHz link LNA. Mini-Circuits PGA-103+ datasheet; image hosted by LCSC.

# Question: Where Should the LNA Go?

Source: 30 K. Cable: 290 K physical. LNA NF: 0.69 dB.




Both give 20 − 3.0 = 17.0 dB signal gain.
Will they give the same output noise?

* Cable loss is exactly half power (3.0 dB). A/B identify arrangements on this slide.

# Worked Answer: Both Component Orders

Friis: $F_{total} = F_1 + (F_2 - 1)/G_1$; $NF_{total} = 10 \log_{10} F_{total}$.

A




Cable first: $F_{total} = 2 + (F_{LNA} - 1)/0.5$; $NF_{total} \approx 3.7$ dB.

B




LNA first: $F_{total} = F_{LNA} + (2 - 1)/100$; $NF_{total} \approx 0.73$ dB.

Combined receiver NF: 3.7 dB with cable first; 0.73 dB with LNA first.
NF improvement ≈ 3.0 dB, using the 290 K reference source.*

* Matched 290 K cable: F = 2, G = 0.5. LNA: F ≈ 1.17, G = 100.

# LNA Placement
# Why Our Link Improves by 7.0 dB

$T_{e,total} = 290(F_{total} - 1)$; $T_{sys} = T_{source} + T_{e,total}$.

Cable first adds 390 K. LNA first adds 52.9 K.

NF compares both arrangements with a 290 K reference source:

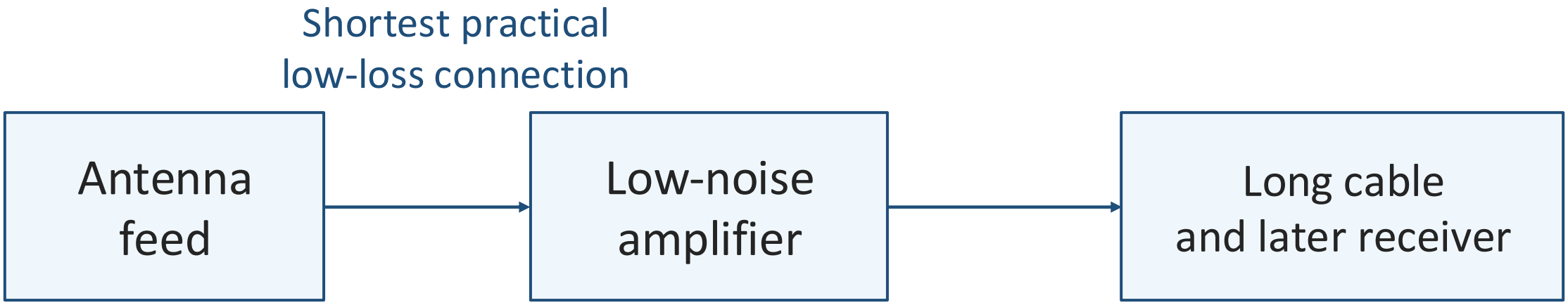
$10 \log_{10}[(290 + 390)/(290 + 52.9)] \approx 3.0$ dB.

Our antenna has only 30 K, so the actual $[S/N_0]$ improves by:

$10 \log_{10}[(30 + 390)/(30 + 52.9)] \approx 7.0$ dB.

* Same signal gain in both arrangements. A quieter antenna makes receiver-added noise more important.

# What It Means for Receiver Layout




Low LNA noise figure sets a low receiver-noise floor.
Enough early gain reduces later input-referred noise.
Check overload, stability, bandwidth, and input filtering.*

* Design guidance: JPL DESCANSO, Vol. 10, Ch. 2; Analog Devices AN-2622.

# Why DSN Uses Cryogenic LNAs



Low-Noise Amplifier (LNA)

A refrigerator cools the LNA.
Physical cooling can lower its NF and receiver-added noise.*

Physical Cooling, Lower Noise

A lower LNA NF reduces the receiver contribution.
The remaining antenna and feed noise limit the improvement in total system sensitivity.

* Cryogenic LNA sources: JPL DESCANSO Ch. 5; S10. DSN photo: NASA/JPL-Caltech, PIA17791.

# Receiving Sensitivity: G/T

G/T is the receiving system's figure of merit.

Combine receive antenna gain with total system noise.

$[T_{sys}] = 10 \log_{10}(T_{sys}/K)$, in dB-K.

$[G/T] = [G_r] - [T_{sys}]$, in dB/K.*

A 1.0 dB gain in [G/T] adds 1.0 dB to $[S/N_0]$.

* Use antenna-system gain and total noise at the same plane. Hold other link terms fixed.

# Worked Example: Cryogenic Cooling

Receiver NF: 0.69 to 0.15 dB.* $T_e = 290(10^{NF/10} - 1)$.

$T_{e,rx}$: about 50 K to 10 K.

$T_{sys,A} = T_A + (L_f - 1)T_{phys} + L_f T_{e,rx}$.

168.0 K (22.3 dB-K) becomes 117.7 K (20.7 dB-K).

$\Delta[G/T] = \Delta[S/N_0] = 10 \log_{10}(168.0/117.7) \approx 1.5$ dB.

* Illustrative receiver specifications. Equivalent noise temperature differs from hardware temperature.

# Question: Which Change Helps?

A. Increase receive antenna gain by 3 dB; hold $T_{sys}$ fixed.

B. Halve total system noise temperature; hold gain fixed.

C. Add 20 dB of ideal gain after the receiver.

Which changes G/T and $S/N_0$? Which changes only output levels?

# Worked Answer: Gain versus Noise

A. Adding 3 dB of antenna gain raises [G/T] and [$S/N_0$] by 3 dB.

B. Halving $T_{sys}$ raises [G/T] and [$S/N_0$] by 3.0 dB.

C. Ideal final gain raises signal and noise equally. The ratio is unchanged.

Sensitivity improves when more signal is collected or less noise is added.

# Example Inputs and Assumptions



384,400 km

Spacecraft → Earth

8.4 GHz return link



1 W transmitter

0.3 m TX dish

34 m receive dish

Aperture efficiency 0.55

TX line: 0.60 dB. $T_A$ = 30 K. Feed: 1 dB at 290 K.

Receiver NF = 0.69 dB ($T_{e,rx}$ ≈ 50 K). Off the lunar disk.*

* Illustrative link assumptions. Images: NASA SVS / LRO; NASA/JPL-Caltech, PIA17791.

# The Tools: Antenna Gain and Path Loss

$G = \eta(\pi Df/c)^2$

For $\eta = 0.55$:

$[G] = 17.8 + 20 \log_{10}(D/m) + 20 \log_{10}(f/GHz)$

$[L_p] = 92.4 + 20 \log_{10}(R/km) + 20 \log_{10}(f/GHz)$

Use diameter in meters and range in kilometers.

# Question: Build the Signal Budget

Let's calculate $[G_{tx}]$, $[G_r]$, and $[L_p]$.

Other losses: 0.1 dB atmosphere + 0.8 dB pointing

+ 0.5 dB polarization = 1.4 dB.

Where do the 0.6 dB TX loss and 1 dB receive-feed loss go?

At terminal A, has the signal passed through the receive feed?

# Worked Answer: Signal Power at A

$$[S_A] = [P_{tx}] - [L_{tx}] + [G_{tx}] - [L_p] - [L_{other}] + [G_r]$$

| | |
|---|---|
| Transmitter power | +0.0 dBW |
| Transmitter line loss | −0.6 dB |
| Transmit antenna gain | +25.8 dBi |
| Free-space loss | −222.6 dB |
| Atmosphere + pointing + polarization | −1.4 dB |
| Receive antenna gain | +66.9 dBi |
| **Received $S_{A*}$** | **−131.9 dBW** |

* Plane A precedes the receive feed. At B, the signal is 1 dB lower.

# Worked Answer: System Noise at A

## All noise referred to antenna terminal A

$L_f = 10^{1.0/10}$; $T_{sys,A} = T_A + (L_f - 1)T_{phys} + L_f T_{e,rx}$.

Feed: $(L_f - 1)290 \approx 75.1$ K; receiver via feed: $L_f(50) \approx 62.9$ K.




$$[T_{sys,A}] = 10 \log_{10}(168.0) = 22.3 \text{ dB-K}.$$

$$[G/T]_A = [G_r] - [T_{sys,A}] \approx 44.7 \text{ dB/K}.$$

# Lunar Link
# Worked Answer: Signal-to-Noise Density

$N_{0,A} = kT_{sys,A}$; $[N_{0,A}] = [k] + [T_{sys,A}]$.

With $[k] = -228.6$: $[N_{0,A}] \approx -206.35$ dBW/Hz.

$[S/N_0] = [S_A] - [N_{0,A}]$.

$= -131.87 - (-206.35) = 74.48 \approx 74.5$ dB-Hz.

# When Bandwidth Is Needed

$N_0$ tells us the noise power per hertz.

Bandwidth B tells us how much noise a receiver or instrument admits.

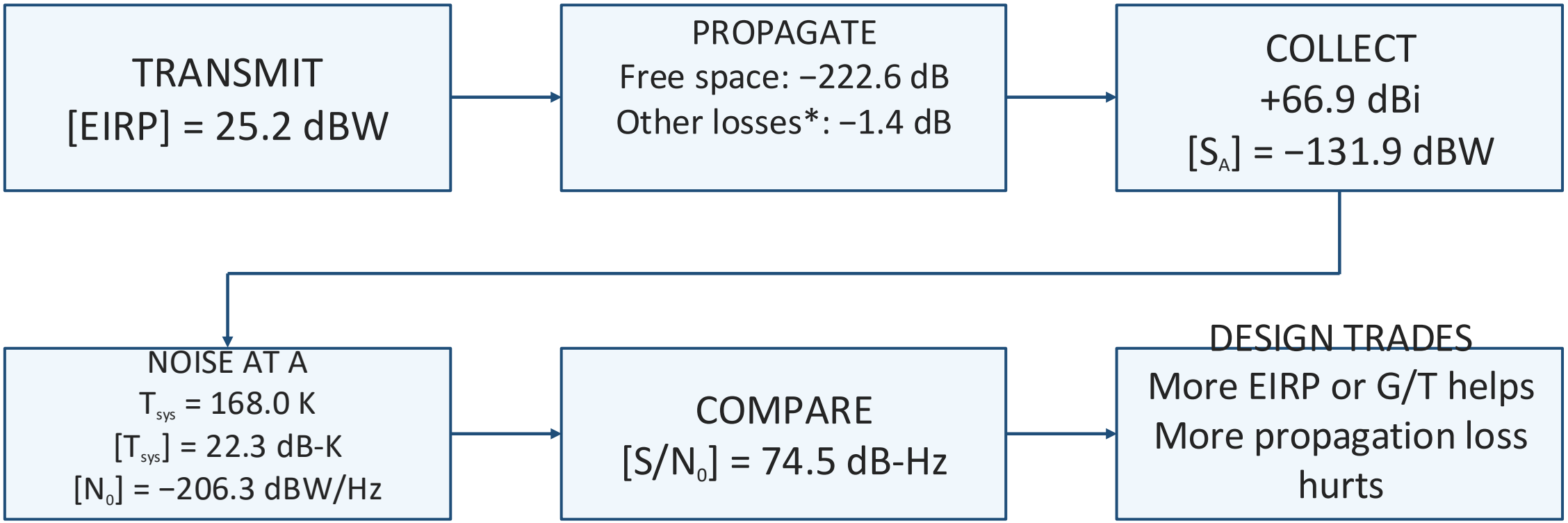
We need total noise $N = N_0B$ to compare measured powers as S/N.

$[S/N] = [S/N_0] - 10 \log_{10}(B/Hz)$.

At 1 MHz: $[S/N] = 74.5 - 60.0 = 14.5$ dB.*

* Narrowing B helps only while passing the needed signal. Use equivalent noise bandwidth.

# The Complete Budget in One View




* Other losses: atmosphere 0.1 dB + pointing 0.8 dB + polarization 0.5 dB.

This is the calculated $S/N_0$. Modulation will determine what value is sufficient.

# What Changes When We Point at the Moon?



## Same Signal Power

$T_A$: 30 K to 230 K

$T_{sys,A}$: 168.0 K to 368.0 K

$[T_{sys,A}]$: 22.3 to 25.7 dB-K*

$T_{sys,new} = T_{sys,old} + (230 - 30) = 368.0$ K.

$\Delta[S/N_0] = -10 \log_{10}(368.0/168.0) = -3.4$ dB.

$[S/N_0]_{new} = 74.5 - 3.4 = 71.1$ dB-Hz.

* Assumed antenna temperatures at fixed signal power. Visible-light Moon image: NASA SVS / LRO.

# Questions: What Changes in the Budget?

1. Double received signal power; hold noise density fixed.
2. Double range; hold antenna gains and temperatures fixed.
3. Double $T_{sys}$; hold the signal budget fixed.
4. Add 0.5 dB atmospheric loss. Does $N_0$ also change?

# Lunar Link
# Worked Answers: Changes in $S/N_0$

Double S: $\Delta[S/N_0] = 10 \log_{10}(2) = +3.0$ dB.

Double range: $\Delta[L_p] = 20 \log_{10}(2) = +6.0$ dB.

Double $T_{sys}$: $\Delta[G/T] = -10 \log_{10}(2) = -3.0$ dB.

Weather: $\Delta[S/N_0] = -\Delta[L_{atm}] + \Delta[G/T]$.*

Update both atmospheric signal loss and antenna noise.

* Other link terms held fixed. Atmospheric emission can change G/T.

# Questions for a Ground-Network Provider

What is G/T at our frequency, elevation, and weather condition?

Which losses and reference plane are included?

What receiver NF and input losses underlie the specification?

What antenna-noise assumptions were used?

# Questions: Putting It All Together

Why should the LNA be close to the antenna?

Which LNA parameters reduce total receiver noise?

What does G/T tell us that receiver NF alone cannot?

Which $S/N_0$ terms can the spacecraft or ground provider change?

How would range, weather, or pointing affect the budget?

# Lesson Takeaways
# LNA Design and Receiving Sensitivity

LNA placement: minimize loss before the first amplifier.

LNA parameters: low NF and enough early gain reduce receiver noise.*

Receiving system: G/T combines antenna gain with all system noise.

Link trades: more EIRP or G/T improves $S/N_0$.

Propagation, pointing, and polarization losses reduce it.

* Gain must also satisfy overload and stability limits. Antenna and feed noise remain.

# The Complete Link Calculation
# The Terms Behind the Trades

$$[S/N_0] = [EIRP] - [L_p] - [L_{atm}] - [L_{point}] - [L_{pol}] + [G/T] + 228.6$$

EIRP: transmitter power, transmit-line loss, and transmit antenna gain.

Losses: free space, atmosphere, pointing, and polarization.

G/T: receive gain and total system noise at one plane.*

$25.2 - 222.6 - 0.1 - 0.8 - 0.5 + 44.7 + 228.6$

$[S/N_0] = 74.5$ dB-Hz

* Receive-feed effects are already in G/T. The constant is −[k] for EIRP in dBW and G/T in dB/K.

# References: Noise and Receiver Models

JPL DESCANSO, Vol. 10, Ch. 2: System Noise Concepts with DSN Applications
NIST: Johnson-noise thermometry and SI defining constants
JPL, Report 42-143: Follow-up Receiver Noise-Temperature Contribution
Keysight: Noise Figure Application / Noise Concepts
JPL DESCANSO, Ch. 5: HEMT Low-Noise Amplifiers
Analog Devices AN-2622: Selecting an RF Low Noise Amplifier

Complete publication links are in the separate Trusted Sources document.

# References and Further Reading

DSN 810-005, Modules 104 and 105: station and environmental noise
ITU-R P.838-3: rain attenuation; P.618-14: Earth-space propagation
Condon & Ransom, Essential Radio Astronomy, Chs. 1–3
Mini-Circuits PGA-103+: manufacturer LNA specifications
ITU-R P.525-5: free-space attenuation
Images: NASA SVS / LRO; NASA/JPL-Caltech; Mini-Circuits via LCSC

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